Risk and Risk Factors

100 Questions: Algorithmic Bias, Fairness, Explainability, GenAI Risks

GARP RAI Exam Preparation

Q1: AI Risks Overview - Tutorial

QUESTION: Why is understanding AI risks particularly challenging?

OPTIONS:

1. AI systems are always perfectly reliable
2. AI is hugely pervasive in range and variety, with rapidly developing capabilities
3. AI risks are all well-understood and documented
4. AI systems never fail

ANSWER: B

DETAILED EXPLANATION:

- AI is pervasive across many domains (finance, healthcare, justice)
- AI capabilities are rapidly developing
- High-stakes applications mean failures have significant consequences
- Risk measurement and mitigation are challenging
- Regulatory responses are often slow to develop

WHY OTHER OPTIONS ARE WRONG:

- A: AI systems are NOT always perfectly reliable
- C: Many AI risks are still being discovered
- D: AI systems can and do fail

EXAM TRAP: AI risks are evolving, pervasive, and high-stakes!

Q2: Pervasiveness Challenge - Tutorial

QUESTION: Why does the pervasiveness of AI pose a risk management challenge?

OPTIONS:

1. AI is only used in one industry
2. Risks take different forms depending on the use case
3. All AI risks are identical
4. Pervasiveness makes risks easier to predict

ANSWER: B

DETAILED EXPLANATION:

- AI applications span many domains (credit scoring, healthcare, policing)
- Each domain has unique risk profiles
- A loan denial algorithm has different risks than a medical diagnostic algorithm
- Risk prediction must account for context

WHY OTHER OPTIONS ARE WRONG:

- A: AI is used across MANY industries
- C: Risks vary significantly by use case
- D: Pervasiveness makes risks HARDER to predict

EXAM TRAP: AI risks are context-dependent across different use cases!

Q3: High-Stakes AI Use - Tutorial

QUESTION: What is a consequence of deploying AI in high-stakes settings?

OPTIONS:

1. Technological shortcomings have minimal impact
2. Technological shortcomings can have significant consequences for people
3. AI is never used in high-stakes settings
4. High-stakes use eliminates all risks

ANSWER: B

DETAILED EXPLANATION:

- Loan applications affect financial well-being
- Medical diagnoses affect health outcomes
- Bail decisions affect personal freedom
- Algorithmic errors can have profound impacts

WHY OTHER OPTIONS ARE WRONG:

- A: High-stakes means impacts are SIGNIFICANT
- C: AI IS used in high-stakes settings
- D: High-stakes use increases, not eliminates, risk

EXAM TRAP: High-stakes AI = significant consequences for individuals!

Q4: Rapid AI Development - Tutorial

QUESTION: What challenge does rapid AI development create?

OPTIONS:

1. It makes risk measurement easier
2. It creates challenges for impact prediction, risk measurement, and governance
3. It eliminates all regulatory concerns
4. It slows down innovation

ANSWER: B

DETAILED EXPLANATION:

- AI capabilities advance faster than our understanding of risks
- Regulatory responses lag behind technological change
- Governance mechanisms are difficult to establish quickly
- Slow regulation allows harmful technologies to persist

WHY OTHER OPTIONS ARE WRONG:

- A: Rapid development makes measurement HARDER
- C: Regulatory concerns are significant
- D: Rapid development ACCELERATES innovation

EXAM TRAP: Rapid AI development = governance and regulation challenges!

Q5: Interconnected AI Risks - Tutorial

QUESTION: How are AI risks typically related?

OPTIONS:

1. They are completely independent of each other
2. They are often interconnected - lack of transparency affects fairness, bias leads to reputational risks
3. There is only one type of AI risk
4. Risks are never interconnected

ANSWER: B

DETAILED EXPLANATION:

- Transparency issues affect fairness assessment
- Algorithmic bias leads to reputational damage
- Safety failures can lead to fairness concerns
- Risks form a complex network

WHY OTHER OPTIONS ARE WRONG:

- A: Risks are highly interconnected
- C: There are many types of AI risks
- D: Risks ARE interconnected

EXAM TRAP: AI risks are interconnected and mutually reinforcing!

Q6: Algorithmic Bias Definition - Tutorial

QUESTION: What is algorithmic bias?

OPTIONS:

1. Random errors in algorithmic predictions
2. A systematic deviation of an algorithm's output, performance, or impact relative to some norm or standard
3. Algorithms that are always perfectly accurate
4. Errors that only occur in supervised learning

ANSWER: B

DETAILED EXPLANATION:

- Systematic (not random) deviation
- Can affect output, performance, or impact
- Relative to some norm, aim, standard, or baseline
- Can be explicit (intentional) or implicit (unintended)

WHY OTHER OPTIONS ARE WRONG:

- A: Bias is SYSTEMATIC, not random
- C: Algorithms CAN be biased
- D: Bias can occur in ANY type of learning

EXAM TRAP: Algorithmic bias = systematic deviation from a norm!

Q7: Explicit vs Implicit Bias - Tutorial

QUESTION: What is the difference between explicit and implicit bias?

OPTIONS:

1. Explicit bias is always worse than implicit bias
2. Explicit bias is intentional/projected by developers; implicit bias creeps in undetected
3. Implicit bias is intentional; explicit bias is accidental
4. There is no difference

ANSWER: B

DETAILED EXPLANATION:

- **Explicit bias:** Developer intentionally projects own biases
- **Implicit bias:** Bias creeps in undetected through design or data
- Both types can lead to unfair outcomes
- Implicit bias is often harder to detect and correct

WHY OTHER OPTIONS ARE WRONG:

- A: Both can be equally harmful
- C: This is reversed
- D: They differ in intentionality

EXAM TRAP: Explicit = intentional; Implicit = unintended!

QUESTION: What is individual fairness in algorithmic design?

OPTIONS:

1. Treating all groups equally
2. The doctrine of "treating like cases alike"
3. Ensuring equal outcomes across demographics
4. Always using protected characteristics

ANSWER: B

DETAILED EXPLANATION:

- Based on Aristotelian doctrine
- Similarly situated individuals should be treated equally
- Focuses on individuals, not groups
- Requires agreement on what counts as "alike"

WHY OTHER OPTIONS ARE WRONG:

- A: This is group fairness
- C: This is demographic parity
- D: Individual fairness avoids using protected characteristics

EXAM TRAP: Individual fairness = treating like cases alike!

Q9: Group Fairness Definition - Tutorial

QUESTION: What is group fairness?

OPTIONS:

1. Treating each individual uniquely
2. Examining statistical differences across groups
3. Only considering individuals with protected characteristics
4. Ignoring group-level patterns

ANSWER: B

DETAILED EXPLANATION:

- Looks at groups sharing protected characteristics
- Asks whether groups are disadvantaged
- Examples: admission rates by gender, loan approvals by race
- Does not consider individuals directly

WHY OTHER OPTIONS ARE WRONG:

- A: This is individual fairness
- C: Group fairness considers ALL groups
- D: Group fairness EXAMINES group patterns

EXAM TRAP: Group fairness = statistical differences across groups!

Q10: College Admission Example - Tutorial

QUESTION: In the college admission example, what fairness issue arises?

OPTIONS:

1. All applicants are treated fairly
2. Individual fairness requires agreeing on what counts as "alike"
3. Socioeconomic factors never matter
4. Algorithms always make perfect decisions

ANSWER: B

DETAILED EXPLANATION:

- What counts as "alike" is often contentious
- Should we consider hardships overcome?
- How to translate resilience into numbers?
- Different people have different views on relevant criteria

WHY OTHER OPTIONS ARE WRONG:

- A: Fairness is not automatically achieved
- C: Socioeconomic factors are often considered
- D: Algorithms can make unfair decisions

EXAM TRAP: Individual fairness requires agreement on what is "alike"!

Q11: Demographic Parity Definition - Tutorial

QUESTION: What is demographic parity?

OPTIONS:

1. Predictions must be equally accurate across groups
2. The distribution of predictions must be identical across subpopulations
3. Positive cases must be correctly identified equally across groups
4. Precision must be equal across groups

ANSWER: B

DETAILED EXPLANATION:

- $P(\hat{Y} = 1 | A = 0) = P(\hat{Y} = 1 | A = 1)$
- Outcome distribution is identical across groups
- Example: same admission rate across genders
- Independent of true values (doesn't consider correctness)

WHY OTHER OPTIONS ARE WRONG:

- A: This is predictive rate parity/equal opportunity
- C: This is equal opportunity
- D: This is predictive rate parity

EXAM TRAP: Demographic parity = equal distribution of predictions!

Q12: Demographic Parity Formula - Tutorial

QUESTION: What is the mathematical definition of demographic parity?

OPTIONS:

1. $P(Y = 1|\hat{Y} = 1, A = 0) = P(Y = 1|\hat{Y} = 1, A = 1)$
2. $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$
3. $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$
4. $P(Y = 1|A = 0) = P(Y = 1|A = 1)$

ANSWER: B

DETAILED EXPLANATION:

- $\hat{Y}$ = predicted outcome
- A = protected attribute/group membership
- Probability of positive prediction given group membership
- Does not depend on actual outcomes (Y)

WHY OTHER OPTIONS ARE WRONG:

- A: This is predictive rate parity (uses precision)
- C: This is equal opportunity (uses sensitivity)
- D: This is base rate equality

EXAM TRAP: Demographic parity = $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$!

Q13: Demographic Parity Drawback - Tutorial

QUESTION: What is a major drawback of demographic parity?

OPTIONS:

1. It is always impossible to achieve
2. It does not consider the quality/accuracy of predictions
3. It is too computationally expensive
4. It requires protected characteristics

ANSWER: B

DETAILED EXPLANATION:

- Independent of true values
- Can be satisfied even if all predictions are wrong
- Example: Unqualified candidates labeled as qualified equally across groups
- Doesn't ensure predictions are correct

WHY OTHER OPTIONS ARE WRONG:

- A: Not always impossible, but can be challenging
- C: It's computationally simple
- D: It does require knowing group membership

EXAM TRAP: Demographic parity ignores prediction accuracy/quality!

QUESTION: What is predictive rate parity?

OPTIONS:

1. The distribution of predictions is identical across groups
2. Precision (TP/(TP+FP)) is equal across groups
3. Sensitivity (TP/(TP+FN)) is equal across groups
4. Accuracy is equal across groups

ANSWER: B

DETAILED EXPLANATION:

- $P(Y = 1|\hat{Y} = 1, A = 0) = P(Y = 1|\hat{Y} = 1, A = 1)$
- Among those predicted positive, proportion correctly identified is same
- Also called "positive predictive value parity"
- Ensures false positive rates are balanced across groups

WHY OTHER OPTIONS ARE WRONG:

- A: This is demographic parity
- C: This is equal opportunity
- D: This is accuracy parity (different concept)

EXAM TRAP: Predictive rate parity = equal precision across groups!

Q15: Predictive Rate Parity Drawback - Tutorial

QUESTION: What is a drawback of predictive rate parity?

OPTIONS:

1. It ignores true positives
2. It does not account for differences in base rates across groups
3. It is always impossible to achieve
4. It requires no data

ANSWER: B

DETAILED EXPLANATION:

- Base rates may differ significantly across groups
- Achieving parity becomes extremely challenging
- Can have adverse effects on algorithmic performance
- May require sacrificing accuracy for fairness

WHY OTHER OPTIONS ARE WRONG:

- A: It specifically considers true positives (precision)
- C: Not always impossible, but can be difficult
- D: Requires data to calculate

EXAM TRAP: Predictive rate parity struggles with different base rates!

Q16: Equal Opportunity Definition - Tutorial

QUESTION: What is equal opportunity?

OPTIONS:

1. The distribution of predictions is identical across groups
2. Precision is equal across groups
3. Sensitivity (TP/(TP+FN)) is equal across groups
4. All predictions are always correct

ANSWER: C

DETAILED EXPLANATION:

- $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$
- True positive rate is equal across groups
- Positive cases equally likely to be identified correctly
- Important in medical diagnostics (don't miss disease)

WHY OTHER OPTIONS ARE WRONG:

- A: This is demographic parity
- B: This is predictive rate parity
- D: No model is always correct

EXAM TRAP: Equal opportunity = equal true positive rate (sensitivity)!

Q17: Equal Opportunity Formula - Tutorial

QUESTION: What is the mathematical definition of equal opportunity?

OPTIONS:

1. $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$
2. $P(Y = 1|\hat{Y} = 1, A = 0) = P(Y = 1|\hat{Y} = 1, A = 1)$
3. $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$
4. $P(Y = 1|A = 0) = P(Y = 1|A = 1)$

ANSWER: C

DETAILED EXPLANATION:

- $\hat{Y}$ = predicted outcome
- Y = actual outcome
- A = protected attribute
- Probability of correct positive prediction given actual positive

WHY OTHER OPTIONS ARE WRONG:

- A: This is demographic parity
- B: This is predictive rate parity
- D: This is base rate equality

EXAM TRAP: Equal opportunity = $P(\hat{Y} = 1|Y = 1, A = 0) = P(\hat{Y} = 1|Y = 1, A = 1)$!

Q18: Equalized Odds Definition - Tutorial

QUESTION: What is equalized odds?

OPTIONS:

1. Equal distribution of predictions
2. Equal precision across groups
3. Equal true positive rate AND equal false positive rate across groups
4. Equal accuracy across groups

ANSWER: C

DETAILED EXPLANATION:

- More restrictive than equal opportunity
- Requires both sensitivity AND specificity equality
- $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$
- $P(\hat{Y} = 1 | Y = 0, A = 0) = P(\hat{Y} = 1 | Y = 0, A = 1)$

WHY OTHER OPTIONS ARE WRONG:

- A: This is demographic parity
- B: This is predictive rate parity
- D: Accuracy parity is different

EXAM TRAP: Equalized odds = equal TPR AND equal FPR!

Q19: Fairness Impossibility Theorem - Tutorial

QUESTION: What does the fairness impossibility theorem state?

OPTIONS:

1. All fairness measures can be satisfied simultaneously
2. It is impossible to satisfy demographic parity, predictive rate parity, and equal opportunity simultaneously when base rates differ
3. Fairness is always achievable
4. Only one fairness measure exists

ANSWER: B

DETAILED EXPLANATION:

- When base rates differ between groups
- Cannot satisfy all three fairness notions
- Trade-offs are unavoidable
- Must deliberate on which fairness measure to prioritize

WHY OTHER OPTIONS ARE WRONG:

- A: This is impossible (the theorem proves it)
- C: Fairness has inherent trade-offs
- D: There are multiple fairness measures

EXAM TRAP: Cannot satisfy all fairness measures simultaneously when base rates differ!

Q20: Accuracy Limitation - Tutorial

QUESTION: Why is accuracy not always a good performance measure?

OPTIONS:

1. It is always a perfect measure
2. It can be misleading with unbalanced data
3. It is too difficult to calculate
4. It only works for regression

ANSWER: B

DETAILED EXPLANATION:

- Example: 100 applicants, 90 unsuitable, 10 suitable
- Algorithm classifies all as unsuitable: Accuracy $= 90\%$
- But algorithm provides no useful information (all unsuitable)
- Especially problematic in unbalanced datasets

WHY OTHER OPTIONS ARE WRONG:

- A: Accuracy has significant limitations
- C: Accuracy is easy to calculate
- D: Accuracy is for classification, not regression

EXAM TRAP: Accuracy is misleading with unbalanced data!

Q21: Precision Definition - Tutorial

QUESTION: What is precision in classification?

OPTIONS:

1. TP/(TP+FN)
2. TP/(TP+FP)
3. (TP+TN)/(TP+TN+FP+FN)
4. TN/(TN+FP)

ANSWER: B

DETAILED EXPLANATION:

- Proportion of positive predictions that are correct
- Also called Positive Predictive Value (PPV)
- $Precision = \frac{TP}{TP+FP}$
- Answers: "Of those predicted positive, how many were actually positive?"

WHY OTHER OPTIONS ARE WRONG:

- A: This is recall/sensitivity
- C: This is accuracy
- D: This is specificity

EXAM TRAP: Precision = TP/(TP+FP) - "of predicted positives, how many were right?"

QUESTION: What is sensitivity (recall) in classification?

OPTIONS:

1. TP/(TP+FN)
2. TP/(TP+FP)
3. (TP+TN)/(TP+TN+FP+FN)
4. TN/(TN+FP)

ANSWER: A

DETAILED EXPLANATION:

- Proportion of actual positives that are correctly identified
- Also called True Positive Rate (TPR)
- $Sensitivity = \frac{TP}{TP+FN}$
- Answers: "Of those actually positive, how many did we catch?"

WHY OTHER OPTIONS ARE WRONG:

- B: This is precision
- C: This is accuracy
- D: This is specificity

EXAM TRAP: Sensitivity = TP/(TP+FN) - "of actual positives, how many did we catch?"

Q23: Precision vs Sensitivity Trade-off - Tutorial

QUESTION: What trade-off exists between precision and sensitivity?

OPTIONS:

1. They always move in the same direction
2. There is often a trade-off; maximizing one can reduce the other
3. They are completely independent
4. Precision is always more important

ANSWER: B

DETAILED EXPLANATION:

- To catch all positives (high sensitivity), may increase false positives (lower precision)
- To avoid false positives (high precision), may miss positives (lower sensitivity)
- Example: Fraud detection - high sensitivity vs. precision
- Choice depends on context

WHY OTHER OPTIONS ARE WRONG:

- A: They often move in opposite directions
- C: They are related through the threshold
- D: Importance depends on context

EXAM TRAP: Precision and sensitivity have a trade-off!

Q24: False Positive Rate Definition - Tutorial

QUESTION: What is the False Positive Rate (FPR)?

OPTIONS:

1. FP/(TP+FP)
2. FP/(FP+TN)
3. FN/(TP+FN)
4. TN/(TN+FP)

ANSWER: B

DETAILED EXPLANATION:

- Proportion of actual negatives that are incorrectly classified as positive
- $FPR = \frac{FP}{FP+TN}$
- Also equals 1 - Specificity
- Important for ROC curves and fairness analysis

WHY OTHER OPTIONS ARE WRONG:

- A: This is 1 - precision, not FPR
- C: This is False Negative Rate
- D: This is specificity

EXAM TRAP: FPR = FP/(FP+TN)!

Q25: Confusion Matrix Components - Tutorial

QUESTION: What are the four components of a confusion matrix?

OPTIONS:

1. TP, TN, FP, FN
2. Mean, Median, Mode, Standard Deviation
3. Training, Validation, Test, Production
4. Precision, Recall, Accuracy, F1

ANSWER: A

DETAILED EXPLANATION:

- **TP (True Positive):** Correctly predicted positive
- **TN (True Negative):** Correctly predicted negative
- **FP (False Positive):** Incorrectly predicted positive
- **FN (False Negative):** Incorrectly predicted negative

WHY OTHER OPTIONS ARE WRONG:

- B: These are statistical measures
- C: These are data splits
- D: These are performance metrics derived from confusion matrix

EXAM TRAP: Confusion matrix = TP, TN, FP, FN!

Q26: Fairness Measures Summary Table - Tutorial

QUESTION: What question does demographic parity answer?

OPTIONS:

1. "Is the likelihood of a true positive prediction the same across groups?"
2. "Is the likelihood of a positive prediction the same across groups?"
3. "Is the likelihood of identifying actual positives as positive the same across groups?"
4. "Are all predictions accurate?"

ANSWER: B

DETAILED EXPLANATION:

- Demographic parity: equal probability of positive prediction
- $P(\hat{Y} = 1|A = 0) = P(\hat{Y} = 1|A = 1)$
- Independent of whether predictions are correct
- Focuses on outcome distribution

WHY OTHER OPTIONS ARE WRONG:

- A: This is predictive rate parity
- C: This is equal opportunity
- D: This is accuracy

EXAM TRAP: Demographic parity = equal positive prediction probability!

QUESTION: What question does equal opportunity answer?

OPTIONS:

1. "Is the likelihood of a positive prediction the same across groups?"
2. "Is the likelihood of a true positive prediction the same across groups?"
3. "Is the likelihood of identifying actual positives as positive the same across groups?"
4. "Is the likelihood of a negative prediction the same across groups?"

ANSWER: C

DETAILED EXPLANATION:

- Equal opportunity: equal true positive rate
- $P(\hat{Y} = 1 | Y = 1, A = 0) = P(\hat{Y} = 1 | Y = 1, A = 1)$
- Ensures actual positives are equally likely to be caught
- Important when missing positives is costly

WHY OTHER OPTIONS ARE WRONG:

- A: This is demographic parity
- B: This is predictive rate parity
- D: This is not a standard fairness measure

EXAM TRAP: Equal opportunity = equal true positive rate!

Q28: Base Rate Definition - Tutorial

QUESTION: What is a base rate?

OPTIONS:

1. The error rate of an algorithm
2. The general prevalence of something within a population
3. The accuracy of predictions
4. The precision of predictions

ANSWER: B

DETAILED EXPLANATION:

- Example: Prevalence of a disease in a population
- Base rates can differ across groups
- Important for fairness analysis
- Different base rates create fairness trade-offs

WHY OTHER OPTIONS ARE WRONG:

- A: Error rate is different
- C: Accuracy is different
- D: Precision is different

EXAM TRAP: Base rate = prevalence in a population!

Q29: Performance vs Fairness Trade-off - Tutorial

QUESTION: Why is balancing performance and fairness challenging?

OPTIONS:

1. Fairness always improves performance
2. Optimizing for fairness can degrade performance and vice versa
3. They are always perfectly aligned
4. Performance is never affected by fairness

ANSWER: B

DETAILED EXPLANATION:

- Example: Fraud detection
- High sensitivity may lower precision
- Fairness constraints may reduce overall accuracy
- Need to balance based on context

WHY OTHER OPTIONS ARE WRONG:

- A: Fairness often TRADES OFF with performance
- C: They are often in tension
- D: Performance IS affected by fairness constraints

EXAM TRAP: There is often a trade-off between performance and fairness!

Q30: Fairness Measures Summary - Tutorial

QUESTION: Which fairness measure is most appropriate for medical diagnostics?

OPTIONS:

1. Demographic parity
2. Predictive rate parity
3. Equal opportunity
4. Accuracy parity

ANSWER: C

DETAILED EXPLANATION:

- Equal opportunity ensures positive cases are caught
- Missing a disease (false negative) can be life-threatening
- Same TPR across groups is critical
- Focuses on sensitivity (not missing positives)

WHY OTHER OPTIONS ARE WRONG:

- A: Demographic parity ignores accuracy
- B: Predictive rate parity focuses on positive predictions
- D: Accuracy may be misleading with imbalanced data

EXAM TRAP: Medical diagnostics prioritizes equal opportunity (don't miss disease)!

Q31: Problem Specification Bias - Tutorial

QUESTION: Why can problem specification introduce bias?

OPTIONS:

1. It always produces perfect algorithms
2. How a problem is specified affects algorithmic performance and fairness
3. Problem specification never affects outcomes
4. All problems are specified identically

ANSWER: B

DETAILED EXPLANATION:

- Choice of criteria affects who applies and who gets hired
- Translation to measurable features requires proxies
- Some concepts are harder to quantify than others
- Example: Productivity vs. teamwork

WHY OTHER OPTIONS ARE WRONG:

- A: Problem specification can introduce bias
- C: Problem specification significantly affects outcomes
- D: Problems vary widely in specification

EXAM TRAP: Problem specification = critical for fairness!

Q32: Proxy Variables - Tutorial

QUESTION: What is a proxy variable in algorithmic design?

OPTIONS:

1. A variable that perfectly measures the desired concept
2. A variable used as a stand-in for a concept that is hard to measure directly
3. A variable that is never used in algorithms
4. A variable with no correlation to the desired concept

ANSWER: B

DETAILED EXPLANATION:

- Used when direct measurement is difficult
- Example: Productivity measured by sales
- Risk: Proxies reduce complex concepts
- Can introduce bias if proxies are imperfect

WHY OTHER OPTIONS ARE WRONG:

- A: Proxies are imperfect by definition
- C: Proxies ARE used in algorithms
- D: Proxies should be correlated with the concept

EXAM TRAP: Proxies are imperfect stand-ins for hard-to-measure concepts!

Q33: Amazon Recruiting Tool Example - Tutorial

QUESTION: What happened with Amazon's recruiting tool?

OPTIONS:

1. It worked perfectly and was deployed
2. It systematically downgraded CVs from women applicants
3. It eliminated all bias
4. It was immediately successful

ANSWER: B

DETAILED EXPLANATION:

- Amazon attempted to create an unbiased recruiting tool
- Algorithm systematically downgraded women's CVs
- Even identical CVs were rated lower for women
- Amazon eventually abandoned the project

WHY OTHER OPTIONS ARE WRONG:

- A: The tool failed due to bias
- C: It perpetuated existing biases
- D: The project was abandoned

EXAM TRAP: Amazon's tool systematically discriminated against women!

Q34: Fairness Through Unawareness - Tutorial

QUESTION: What is "fairness through unawareness"?

OPTIONS:

1. Using all available features
2. Removing protected characteristics from input features
3. Always including protected characteristics
4. Ignoring all features

ANSWER: B

DETAILED EXPLANATION:

- Intuitive approach: don't use protected characteristics
- Assumes removing them prevents discrimination
- Problem: correlated variables can still carry protected information
- Example: Prior convictions correlated with protected group membership

WHY OTHER OPTIONS ARE WRONG:

- A: This would increase bias risk
- C: This would allow explicit discrimination
- D: This would make predictions impossible

EXAM TRAP: Fairness through unawareness does NOT eliminate bias!

Q35: FTA Prediction Example - Tutorial

QUESTION: In the FTA prediction example, what happened when protected characteristics were removed?

OPTIONS:

1. Bias was completely eliminated
2. The coefficient on prior convictions increased from 54
3. The algorithm became perfectly fair
4. No changes occurred

ANSWER: B

DETAILED EXPLANATION:

- Protected and prior convictions were correlated
- Removing protected variable contaminated the prior variable
- Prior conviction now carried more weight
- Algorithm still indirectly discriminated

WHY OTHER OPTIONS ARE WRONG:

- A: Bias was NOT eliminated
- C: The algorithm remained biased
- D: Significant changes occurred

EXAM TRAP: Removing protected characteristics can make bias worse through correlated variables!

Q36: Historical Bias - Tutorial

QUESTION: What is historical bias in AI?

OPTIONS:

1. Bias that occurs only in the present
2. Bias in training data that reflects past discrimination
3. Bias that never affects algorithms
4. Bias that is always intentional

ANSWER: B

DETAILED EXPLANATION:

- Supervised algorithms learn from historical data
- Historical data reflects past biases and discrimination
- Example: Women historically less likely selected for roles
- Algorithm repeats and may exacerbate patterns

WHY OTHER OPTIONS ARE WRONG:

- A: Historical bias is about the PAST
- C: Historical bias DOES affect algorithms
- D: Historical bias is often unintentional

EXAM TRAP: Historical bias = past discrimination reflected in training data!

Q37: Feedback Loops - Tutorial

QUESTION: What is a feedback loop in AI systems?

OPTIONS:

1. A system that corrects its own errors
2. When algorithmic predictions influence future data, perpetuating bias
3. A system that never changes
4. A system that eliminates all bias

ANSWER: B

DETAILED EXPLANATION:

- Algorithm's outputs influence future data
- Biased predictions create more biased data
- Amplifies existing discrimination
- Example: Hiring algorithm learns from its own biased outcomes

WHY OTHER OPTIONS ARE WRONG:

- A: Feedback loops often AMPLIFY errors
- C: Feedback loops cause change (often negative)
- D: Feedback loops can perpetuate bias

EXAM TRAP: Feedback loops amplify and perpetuate bias!

Q38: Representation Bias - Tutorial

QUESTION: What is representation bias?

OPTIONS:

1. When data is perfectly representative
2. When the collected sample is not representative of the target population
3. When all groups are equally represented
4. When data collection is unbiased

ANSWER: B

DETAILED EXPLANATION:

- Sample does not reflect target population
- Results in problematic model specifications
- Example: Facial recognition trained mostly on white faces
- Algorithm performs better on dominant groups

WHY OTHER OPTIONS ARE WRONG:

- A: Representation bias means NOT representative
- C: This would be balanced representation
- D: Representation bias IS a bias

EXAM TRAP: Representation bias = non-representative sample!

QUESTION: What is measurement bias?

OPTIONS:

1. When measurement methods are consistent across the sample
2. When measurement methods are not consistent across the sample
3. When all measurements are perfectly accurate
4. When measurements are always reliable

ANSWER: B

DETAILED EXPLANATION:

- Inconsistent measurement methods
- Can favor certain groups
- Example: Doctor spends more time with wealthy patients
- Data on early-stage disease discovery may be biased

WHY OTHER OPTIONS ARE WRONG:

- A: Measurement bias means INCONSISTENT methods
- C: Measurement bias means NOT perfectly accurate
- D: Measurement bias means UNRELIABLE across groups

EXAM TRAP: Measurement bias = inconsistent measurement across groups!

Q40: Data Composition Bias - Tutorial

QUESTION: What is data composition bias?

OPTIONS:

1. When datasets are perfectly balanced
2. When minority subgroups are underrepresented, causing algorithms to underperform
3. When all groups are equally represented
4. When data is always complete

ANSWER: B

DETAILED EXPLANATION:

- Statistical minorities have fewer data points
- Algorithm underperforms for underrepresented groups
- Example: 3
- Algorithm may significantly underperform for this group

WHY OTHER OPTIONS ARE WRONG:

- A: Composition bias means IMBALANCED data
- C: This would be balanced data
- D: Composition bias means INCOMPLETE representation

EXAM TRAP: Data composition bias = underrepresented minority subgroups!

Q41: Balancing Datasets - Tutorial

QUESTION: What does balancing a dataset mean?

OPTIONS:

1. Using proportional representation
2. Ensuring relevant minority subgroups occupy a similar proportion as majority groups
3. Removing all minority data
4. Always using the original distribution

ANSWER: B

DETAILED EXPLANATION:

- Minority groups represented more than proportionally
- Ensures algorithm recognizes minority groups
- Different from proportional representation
- Example: 50

WHY OTHER OPTIONS ARE WRONG:

- A: Proportional representation is different
- C: Balancing increases minority representation
- D: Balancing changes the original distribution

EXAM TRAP: Balancing = over-representing minorities for algorithm performance!

Q42: Bootstrapping Methods - Tutorial

QUESTION: What are methods for balancing datasets?

OPTIONS:

1. Over-sampling minority groups
2. Under-sampling majority groups
3. Both A and B
4. Only removing outliers

ANSWER: C

DETAILED EXPLANATION:

- **Over-sampling:** Add more data from minority group
- **Under-sampling:** Remove data from majority group
- **Combination:** Use both methods
- Can also create synthetic data points

WHY OTHER OPTIONS ARE WRONG:

- A: Correct but incomplete
- B: Correct but incomplete
- D: Outlier removal is different

EXAM TRAP: Bootstrapping methods include over-sampling AND under-sampling!

Q43: Intersectional Bias - Tutorial

QUESTION: What is intersectional bias?

OPTIONS:

1. Bias that only affects one group
2. Bias affecting groups at the intersection of multiple underrepresented subgroups
3. Bias that never occurs
4. Bias that is always intentional

ANSWER: B

DETAILED EXPLANATION:

- Performance can be particularly low for intersectional groups
- Example: Women of color in facial recognition
- Multiple dimensions of underrepresentation
- Harder to detect and correct

WHY OTHER OPTIONS ARE WRONG:

- A: Intersectional bias affects MULTIPLE groups
- C: Intersectional bias DOES occur
- D: Intersectional bias is often unintentional

EXAM TRAP: Intersectional bias = multiple underrepresented dimensions combined!

Q44: Word Embeddings Bias - Tutorial

QUESTION: How can word embeddings introduce bias?

OPTIONS:

1. They always produce unbiased results
2. They can contain stereotypes (e.g., man:computer programmer as woman:homemaker)
3. They never contain stereotypes
4. They are always perfectly neutral

ANSWER: B

DETAILED EXPLANATION:

- Word embeddings capture meaning from text
- Text contains societal stereotypes
- Example: "man is to computer programmer as woman is to homemaker"
- NLP algorithms can incorporate these stereotypes

WHY OTHER OPTIONS ARE WRONG:

- A: Word embeddings CAN contain bias
- C: Word embeddings DO contain stereotypes
- D: Word embeddings reflect societal biases

EXAM TRAP: Word embeddings can contain harmful stereotypes!

Q45: Objective Function Bias - Tutorial

QUESTION: How can objective functions introduce bias?

OPTIONS:

1. They never affect fairness
2. What the algorithm optimizes for makes an ethically significant difference
3. All objective functions are equally fair
4. Objective functions always eliminate bias

ANSWER: B

DETAILED EXPLANATION:

- Focus on performance vs. error distribution
- Cancer detection example: err on side of caution
- Brain cancer: invasive treatment considerations
- Choice depends on context

WHY OTHER OPTIONS ARE WRONG:

- A: Objective functions significantly affect fairness
- C: Different objectives have different fairness implications
- D: Objective functions CAN introduce bias

EXAM TRAP: Objective function choice = ethical significance!

Q46: Cancer Detection Example - Tutorial

QUESTION: In the cancer detection example, why might different algorithms require different approaches?

OPTIONS:

1. All cancers are the same
2. Skin cancer and brain cancer have different treatment risks and consequences
3. Treatment is never invasive
4. All patients are identical

ANSWER: B

DETAILED EXPLANATION:

- Skin cancer: biopsy, relatively low risk
- Brain cancer: major surgery, high risk
- Different trade-offs for false positives/negatives
- Context matters for objective function

WHY OTHER OPTIONS ARE WRONG:

- A: Different cancers have different risks
- C: Some treatments ARE highly invasive
- D: Patients have different conditions

EXAM TRAP: Context determines appropriate trade-offs!

QUESTION: How can benchmark datasets introduce bias?

OPTIONS:

1. They are always representative
2. They may not be representative of the intended use population
3. They always eliminate bias
4. They never affect model performance

ANSWER: B

DETAILED EXPLANATION:

- Public benchmarks may not reflect real-world population
- Good performance on benchmark doesn't guarantee real-world performance
- Can mask biases that appear in deployment
- Need domain-specific testing

WHY OTHER OPTIONS ARE WRONG:

- A: Benchmarks may NOT be representative
- C: Benchmarks can CONTAIN bias
- D: Benchmarks significantly affect perceived performance

EXAM TRAP: Benchmark datasets may not reflect real-world populations!

Q48: Deployment Context Bias - Tutorial

QUESTION: How can deployment context introduce bias?

OPTIONS:

1. Algorithms always perform equally in all contexts
2. Algorithms used outside their designed context can lead to performance loss
3. Context never matters
4. All contexts are identical

ANSWER: B

DETAILED EXPLANATION:

- Algorithm for one context may not work in another
- Example: Recidivism prediction for sentence length
- Historical data may be outdated
- Values of users may differ from developers

WHY OTHER OPTIONS ARE WRONG:

- A: Performance varies by context
- C: Context significantly matters
- D: Contexts vary widely

EXAM TRAP: Deployment context = critical for performance and fairness!

Q49: Human Oversight Importance - Tutorial

QUESTION: Why is human oversight important for AI systems?

OPTIONS:

1. Humans are never needed
2. Ensuring safe and fair algorithms requires human oversight throughout
3. AI systems are always perfectly safe
4. Oversight is optional

ANSWER: B

DETAILED EXPLANATION:

- Data collection process needs oversight
- Model development needs review
- Deployment needs monitoring
- Regular auditing becomes important
- Consequential domains require accountability

WHY OTHER OPTIONS ARE WRONG:

- A: Human oversight is ESSENTIAL
- C: AI systems are NOT always perfectly safe
- D: Oversight is CRITICAL, not optional

EXAM TRAP: Human oversight is essential throughout the AI lifecycle!

Q50: Sources of Unfairness Summary - Tutorial

QUESTION: Which is NOT a source of algorithmic unfairness?

OPTIONS:

1. Problem specification and feature selection
2. Data collection and composition
3. Model development
4. Always using perfect data

ANSWER: D

DETAILED EXPLANATION:

- Problem specification: choice of criteria and proxies
- Data collection: representation and measurement bias
- Data composition: imbalanced datasets
- Model development: embeddings, objective functions, testing
- Deployment: context, outdated data, user values

WHY OTHER OPTIONS ARE WRONG:

- A: This IS a source of unfairness
- B: This IS a source of unfairness
- C: This IS a source of unfairness

EXAM TRAP: Perfect data does NOT exist - it's a source of unfairness!

Q51: Explainability Definition - Tutorial

QUESTION: What is explainability in AI?

OPTIONS:

1. The ability to make predictions
2. The capacity to explain in understandable terms how an AI system makes decisions
3. The speed of the algorithm
4. The accuracy of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- Ex-post explainability (after the fact)
- Explaining decisions in understandable terms
- Essential for accountability and trust
- Particularly important in high-stakes domains

WHY OTHER OPTIONS ARE WRONG:

- A: This is prediction capability
- C: Speed is different
- D: Accuracy is different

EXAM TRAP: Explainability = explaining decisions after the fact!

Q52: Interpretability Definition - Tutorial

QUESTION: What is interpretability in AI?

OPTIONS:

1. Explaining decisions after the fact
2. The degree to which a human can comprehend and predict the model's behavior inherently
3. The speed of the algorithm
4. The accuracy of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- Built-in mechanisms to understand inputs $\longrightarrow$ outputs
- Inherently interpretable models
- Example: Decision trees (can follow the logic)
- Before the fact (ex-ante), not after

WHY OTHER OPTIONS ARE WRONG:

- A: This is explainability
- C: Speed is different
- D: Accuracy is different

EXAM TRAP: Interpretability = inherently understandable models!

Q53: Transparency Definition - Tutorial

QUESTION: What is transparency in AI?

OPTIONS:

1. Making predictions quickly
2. The openness and accessibility of an AI system's design, algorithms, data, and decision-making processes
3. The accuracy of the algorithm
4. The speed of the algorithm

ANSWER: B

DETAILED EXPLANATION:

- Openness about design and algorithms
- Accessibility of data and processes
- Builds trust and enables oversight
- Facilitates collaboration

WHY OTHER OPTIONS ARE WRONG:

- A: Speed is different
- C: Accuracy is different
- D: Speed is different

EXAM TRAP: Transparency = openness and accessibility of AI systems!

Q54: Black-Box Problem - Tutorial

QUESTION: What is the black-box problem in AI?

OPTIONS:

1. Models that are always transparent
2. Models that are too complex to be interpreted by humans
3. Models that are always simple
4. Models that are always accurate

ANSWER: B

DETAILED EXPLANATION:

- Neural networks are often black boxes
- Cannot understand intrinsic architecture
- Leads to explainability issues
- Particularly challenging for high-stakes decisions

WHY OTHER OPTIONS ARE WRONG:

- A: Black boxes are NOT transparent
- C: Black boxes are COMPLEX
- D: Black boxes may NOT be accurate

EXAM TRAP: Black-box problem = models too complex to interpret!

Q55: Feature Importance Scores - Tutorial

QUESTION: What are feature importance scores used for?

OPTIONS:

1. To make predictions faster
2. To rank the importance of input features for outcomes
3. To reduce the number of data points
4. To increase model complexity

ANSWER: B

DETAILED EXPLANATION:

- Used in Explainable AI (XAI)
- Reveals which factors have most influence
- Example: Credit scoring - income vs. credit history
- Helps identify potential biases

WHY OTHER OPTIONS ARE WRONG:

- A: Feature importance is about interpretation
- C: It doesn't reduce data points
- D: It explains, not increases complexity

EXAM TRAP: Feature importance = ranks feature influence!

Q56: Shapley Values - Tutorial

QUESTION: What do Shapley values calculate?

OPTIONS:

1. The accuracy of a model
2. The contribution of each input feature to the model's output
3. The speed of the model
4. The size of the model

ANSWER: B

DETAILED EXPLANATION:

- Calculate marginal effect of including each feature
- Across all possible feature combinations
- Fair attribution of contributions
- Based on game theory (Shapley values)

WHY OTHER OPTIONS ARE WRONG:

- A: Accuracy is different
- C: Speed is different
- D: Size is different

EXAM TRAP: Shapley values = feature contribution calculation!

Q57: LUCID - Tutorial

QUESTION: What does LUCID do in XAI?

OPTIONS:

1. Makes predictions faster
2. Works backward from algorithmic output to reveal internal logic
3. Increases model complexity
4. Removes features

ANSWER: B

DETAILED EXPLANATION:

- LUCID = Locating Unfairness through Canonical Inverse Design
- Works backward from output to input
- Creates distribution over input space
- Reveals model's internal logic

WHY OTHER OPTIONS ARE WRONG:

- A: Speed is not LUCID's purpose
- C: LUCID helps understand complexity
- D: LUCID doesn't remove features

EXAM TRAP: LUCID works backward from output to reveal logic!

QUESTION: What are surrogate models?

OPTIONS:

1. More complex models that replace simpler ones
2. Simpler, more interpretable models that approximate complex AI systems
3. Models that are always less accurate
4. Models that cannot be interpreted

ANSWER: B

DETAILED EXPLANATION:

- Example: LIME uses surrogate models
- Approximate complex model's predictions locally
- Use interpretable models (linear regression, decision trees)
- Enhance interpretability

WHY OTHER OPTIONS ARE WRONG:

- A: Surrogate models are SIMPLER
- C: Surrogate models can be quite accurate locally
- D: Surrogate models are INTERPRETABLE

EXAM TRAP: Surrogate models are simpler, interpretable approximations!

Q59: LIME Definition - Tutorial

QUESTION: What is LIME in XAI?

OPTIONS:

1. Local Interpretable Model-agnostic Explanations
2. A method to make models faster
3. A technique to increase accuracy
4. A method to remove bias

ANSWER: A

DETAILED EXPLANATION:

- LIME = Local Interpretable Model-agnostic Explanations
- Approximates complex model locally
- Uses interpretable model (linear regression, decision tree)
- Model-agnostic (works with any model)

WHY OTHER OPTIONS ARE WRONG:

- B: LIME is about explanation, not speed
- C: LIME is about interpretability, not accuracy
- D: LIME helps identify bias, doesn't remove it directly

EXAM TRAP: LIME = Local Interpretable Model-agnostic Explanations!

QUESTION: What is a challenge with XAI techniques?

OPTIONS:

1. They always produce perfect explanations
2. Oversimplification can distort understanding
3. They are never computationally expensive
4. They always eliminate bias

ANSWER: B

DETAILED EXPLANATION:

- Complex decisions oversimplified
- Can lead to misinterpretation
- Computationally intensive
- Ex-post techniques (after training)
- Trade-off: complex models vs. interpretability

WHY OTHER OPTIONS ARE WRONG:

- A: Explanations can be imperfect
- C: XAI CAN be computationally expensive
- D: XAI helps identify, not eliminate bias

EXAM TRAP: XAI can oversimplify and distort understanding!

Q61: Secrecy - Tutorial

QUESTION: What is secrecy as a form of opaqueness?

OPTIONS:

1. When algorithms are fully transparent
2. When individuals are not aware of being subject to algorithmic decision making
3. When algorithms are always explained
4. When all data is public

ANSWER: B

DETAILED EXPLANATION:

- Example: Algorithm extracts data without knowledge
- Used for targeted advertising
- Individuals unaware of algorithmic profiling
- Violates transparency and consent

WHY OTHER OPTIONS ARE WRONG:

- A: Secrecy is the OPPOSITE of transparent
- C: Secrecy means NOT explained
- D: Secrecy means NOT public

EXAM TRAP: Secrecy = unawareness of algorithmic decision making!

Q62: Confidentiality - Tutorial

QUESTION: What is confidentiality as a form of opaqueness?

OPTIONS:

1. When algorithms are completely transparent
2. When the algorithmic decision-making process is opaque to affected parties
3. When all data is public
4. When algorithms are always explained

ANSWER: B

DETAILED EXPLANATION:

- Individuals know they're subject to AI
- But don't know how decisions are made
- Main concern in AI discourse
- Similar to non-algorithmic secret decision-making

WHY OTHER OPTIONS ARE WRONG:

- A: Confidentiality means NOT transparent
- C: Confidentiality means NOT public
- D: Confidentiality means NOT explained

EXAM TRAP: Confidentiality = opaque decision-making process!

Q63: Specialized Knowledge - Tutorial

QUESTION: What is specialized knowledge as a form of opaqueness?

OPTIONS:

1. When everyone can understand algorithms
2. When understanding algorithms requires specialized training
3. When algorithms are always simple
4. When no knowledge is needed

ANSWER: B

DETAILED EXPLANATION:

- Even with access to source code
- May not be able to make sense of it
- Requires specialized training
- Not unique to AI systems

WHY OTHER OPTIONS ARE WRONG:

- A: Specialized knowledge means NOT everyone understands
- C: Algorithms may be complex
- D: Knowledge IS needed

EXAM TRAP: Specialized knowledge = requires training to understand!

Q64: Reasons for Opacity - Tutorial

QUESTION: What is a valid reason for algorithmic opacity?

OPTIONS:

1. To hide discrimination
2. Security (spam filters) and proprietary interests
3. To avoid accountability
4. To confuse users

ANSWER: B

DETAILED EXPLANATION:

- Security: making code public enables circumvention
- Example: Spam filters would be easier to bypass
- Proprietary interests: trade secrets
- Must balance transparency with legitimate interests

WHY OTHER OPTIONS ARE WRONG:

- A: Hiding discrimination is NOT valid
- C: Avoiding accountability is NOT valid
- D: Confusing users is NOT valid

EXAM TRAP: Legitimate reasons include security and proprietary interests!

Q65: Explainability Summary - Tutorial

QUESTION: What is the most comprehensive statement about explainability?

OPTIONS:

1. Explainability is always easy to achieve
2. Explainability, interpretability, and transparency are essential for accountability and trust in AI systems
3. Explainability is never needed
4. Only black-box models are used

ANSWER: B

DETAILED EXPLANATION:

- **Explainability:** Ex-post understanding of decisions
- **Interpretability:** Inherent understanding of model behavior
- **Transparency:** Openness about design and processes
- All essential for accountability and trust

WHY OTHER OPTIONS ARE WRONG:

- A: Explainability can be challenging
- C: Explainability IS needed for accountability
- D: Many models are interpretable

EXAM TRAP: All three are essential for accountability and trust!

Q66: Human Autonomy Definition - Tutorial

QUESTION: What is human autonomy in the context of AI?

OPTIONS:

1. The ability to make any decision
2. The ability to hold beliefs, make, and execute decisions of our own
3. The ability to control AI systems
4. The ability to avoid all technology

ANSWER: B

DETAILED EXPLANATION:

- Two dimensions:
- **Internal:** Values, beliefs, decisions that are our own
- **External:** Freedom and opportunity to execute decisions
- Not automatically lost by delegating tasks
- Requires meaningful human oversight

WHY OTHER OPTIONS ARE WRONG:

- A: Autonomy is more than making any decision
- C: This is control, not autonomy
- D: Avoiding technology is not autonomy

EXAM TRAP: Autonomy = own beliefs and decisions + ability to execute!

Q67: Cambridge Analytica Scandal - Tutorial

QUESTION: What did the Cambridge Analytica scandal demonstrate?

OPTIONS:

1. The benefits of data collection
2. How big data could be used to manipulate voter behavior
3. That data privacy is not important
4. That AI never affects elections

ANSWER: B

DETAILED EXPLANATION:

- Collected Facebook user data
- Inferred personality profiles from online behavior
- Targeted political advertising
- Demonstrated potential for manipulation

WHY OTHER OPTIONS ARE WRONG:

- A: The scandal showed MISUSE of data
- C: The scandal highlighted privacy concerns
- D: AI CAN affect elections

EXAM TRAP: Cambridge Analytica = political manipulation through data!

Q68: Facebook Emotional Contagion Experiment - Tutorial

QUESTION: What did the Facebook emotional contagion experiment show?

OPTIONS:

1. That emotions cannot be transmitted online
2. That recommendation algorithms can shape and manipulate user experience
3. That users always consent to experiments
4. That AI has no effect on emotions

ANSWER: B

DETAILED EXPLANATION:

- Researchers changed users' timelines
- Displayed predominantly positive or negative posts
- Users' own posts became more positive/negative
- No consent from users
- Caused public outcry

WHY OTHER OPTIONS ARE WRONG:

- A: Emotions CAN be transmitted online
- C: Users did NOT consent
- D: AI CAN affect emotions

EXAM TRAP: Experiment showed AI can manipulate emotional state!

Q69: AI Manipulation Prevention - Tutorial

QUESTION: How can AI manipulation be prevented?

OPTIONS:

1. By ignoring all risks
2. Through human oversight and regulatory oversight
3. By eliminating all AI
4. By never collecting data

ANSWER: B

DETAILED EXPLANATION:

- Human oversight throughout development
- Regulatory oversight (e.g., EU AI Act)
- Transparency obligations
- Prohibiting subliminal techniques
- Balancing benefits vs. risks

WHY OTHER OPTIONS ARE WRONG:

- A: Ignoring risks is not prevention
- C: Eliminating AI is unrealistic
- D: Some data collection is necessary

EXAM TRAP: Prevention requires human and regulatory oversight!

Q70: Autonomy and Manipulation Summary - Tutorial

QUESTION: What is the most comprehensive statement about autonomy and manipulation?

OPTIONS:

1. AI never affects autonomy
2. AI systems can shape our experience and potentially manipulate us, requiring oversight and regulation
3. Manipulation is always beneficial
4. Autonomy is not important

ANSWER: B

DETAILED EXPLANATION:

- AI shapes online experience
- Can influence behavior through recommendation algorithms
- Examples: Cambridge Analytica, emotional contagion
- Requires oversight and regulation

WHY OTHER OPTIONS ARE WRONG:

- A: AI CAN affect autonomy
- C: Manipulation is generally harmful
- D: Autonomy IS important

EXAM TRAP: AI can shape experience and manipulate - requires oversight!

Q71: Uber Crash Example - Tutorial

QUESTION: What did the Uber driverless vehicle crash demonstrate?

OPTIONS:

1. That autonomous vehicles are always safe
2. The risks of over-dependence on AI in critical systems
3. That safety mechanisms are never needed
4. That AI never fails

ANSWER: B

DETAILED EXPLANATION:

- Vehicle's systems failed to recognize pedestrian
- Safety mechanisms were deactivated
- Human safety driver was distracted
- Result of automation bias
- Need for robust testing and validation

WHY OTHER OPTIONS ARE WRONG:

- A: The crash showed they are NOT always safe
- C: Safety mechanisms ARE needed
- D: AI CAN fail

EXAM TRAP: Uber crash = risks of over-reliance on AI!

Q72: Automation Bias Definition - Tutorial

QUESTION: What is automation bias?

OPTIONS:

1. The tendency to distrust automated systems
2. The tendency of humans to over-rely on automated systems
3. The tendency to always use manual methods
4. The tendency to ignore technology

ANSWER: B

DETAILED EXPLANATION:

- Over-reliance on automation
- Leads to complacency and reduced vigilance
- Errors of oversight
- Particularly problematic in critical situations

WHY OTHER OPTIONS ARE WRONG:

- A: This is distrust, not over-reliance
- C: Automation bias is about using automation
- D: Automation bias is about over-relying

EXAM TRAP: Automation bias = over-reliance on automated systems!

QUESTION: What is skill atrophy in the context of AI?

OPTIONS:

1. Improving skills through AI use
2. Decay in essential skills from prolonged reliance on automation
3. Learning new skills quickly
4. Maintaining all skills

ANSWER: B

DETAILED EXPLANATION:

- Operators become passive observers
- Move from active decision makers
- Essential skills deteriorate
- Result of automation bias

WHY OTHER OPTIONS ARE WRONG:

- A: This is skill improvement
- C: Skill atrophy is about DECAY, not learning
- D: Skill atrophy is about LOSING skills

EXAM TRAP: Skill atrophy = decay from over-reliance on automation!

QUESTION: How can automation bias be mitigated?

OPTIONS:

1. By removing all automation
2. By informing users and requiring periodic human input
3. By ignoring the problem
4. By increasing automation

ANSWER: B

DETAILED EXPLANATION:

- Inform users/operators of the risks
- Ensure understanding of limitations
- Implement design mechanisms requiring periodic human input
- Example: Eye tracking technology with alerts

WHY OTHER OPTIONS ARE WRONG:

- A: Removing all automation is unrealistic
- C: Ignoring the problem is not mitigation
- D: Increasing automation worsens the problem

EXAM TRAP: Mitigation = user education + periodic human verification!

Q75: Mental Health AI Risks - Tutorial

QUESTION: What is a concern with AI-driven mental health support?

OPTIONS:

1. AI systems have perfect empathy
2. AI systems have no empathy, which is crucial for mental health support
3. AI systems are always better than humans
4. AI systems never fail in mental health

ANSWER: B

DETAILED EXPLANATION:

- AI can give seemingly empathetic responses
- But AI has no actual empathy
- Positive effects dwindle when patients know it's AI-generated
- Some issues should remain in human hands

WHY OTHER OPTIONS ARE WRONG:

- A: AI does NOT have perfect empathy
- C: AI is NOT always better than humans
- D: AI CAN fail in mental health

EXAM TRAP: AI lacks genuine empathy - critical for mental health!

Q76: Reputational Risk Causes - Tutorial

QUESTION: What are the three main causes of AI-related reputational damage?

OPTIONS:

1. Speed, cost, and accuracy
2. Privacy breaches, algorithmic bias, and lack of explainability
3. Size, complexity, and speed
4. Training data, model type, and deployment

ANSWER: B

DETAILED EXPLANATION:

- **Privacy breaches:** Cambridge Analytica example
- **Algorithmic bias:** Dutch child benefits scandal
- **Lack of explainability:** Customers denied loans without reasons

WHY OTHER OPTIONS ARE WRONG:

- A: Speed, cost, accuracy are not the main causes
- C: Size, complexity, speed are not the main causes
- D: Training data, model type, deployment are not the main causes

EXAM TRAP: Three main causes = privacy, bias, explainability!

Q77: Dutch Child Benefits Scandal - Tutorial

QUESTION: What happened in the Dutch child benefits scandal?

OPTIONS:

1. An algorithm correctly identified fraud
2. A fraud-detection algorithm falsely denied benefits to thousands of citizens
3. All claims were approved
4. The algorithm worked perfectly

ANSWER: B

DETAILED EXPLANATION:

- Algorithm deployed by Dutch government
- Falsely denied and reclaimed child benefits
- Thousands of citizens affected
- Dire consequences for families
- Dutch prime minister resigned

WHY OTHER OPTIONS ARE WRONG:

- A: The algorithm incorrectly identified fraud
- C: Many claims were DENIED
- D: The algorithm failed catastrophically

EXAM TRAP: Dutch scandal = algorithmic bias with severe consequences!

Q78: Competence Criticism - Tutorial

QUESTION: What is competence criticism of companies?

OPTIONS:

1. Criticism that the company lacks relevant expertise
2. Criticism that the company has misaligned values
3. Criticism that the company is too successful
4. Criticism that the company is too transparent

ANSWER: A

DETAILED EXPLANATION:

- Perception that company lacks expertise
- Insufficient capacities for safe/fair algorithms
- Technical failings, unfairness, or lack of explainability
- Data not stored safely

WHY OTHER OPTIONS ARE WRONG:

- B: This is value alignment criticism
- C: This is not criticism
- D: Transparency is generally positive

EXAM TRAP: Competence = lack of expertise or capacity!

Q79: Value Alignment Criticism - Tutorial

QUESTION: What is value alignment criticism of companies?

OPTIONS:

1. Criticism that the company lacks relevant expertise
2. Criticism that the company has misaligned values or acted despite knowing risks
3. Criticism that the company is too successful
4. Criticism that the company is too transparent

ANSWER: B

DETAILED EXPLANATION:

- Company fully aware of risks but proceeded
- Example: Facebook emotional contagion experiment
- Lack of due diligence
- Violation of user trust and expectations

WHY OTHER OPTIONS ARE WRONG:

- A: This is competence criticism
- C: This is not criticism
- D: Transparency is generally positive

EXAM TRAP: Value alignment = misaligned values or knowingly proceeding with risks!

Q80: Reputational Risk Mitigation - Tutorial

QUESTION: What is NOT a reputational risk mitigation strategy?

OPTIONS:

1. Continuous monitoring and evaluation
2. Ignoring incidents and hoping they go away
3. Ethical AI frameworks
4. Stakeholder engagement and communication

ANSWER: B

DETAILED EXPLANATION:

- **Monitoring:** Regular risk assessment
- **Transparency:** Explainable algorithms, user awareness
- **Ethical frameworks:** Guidelines and oversight
- **Stakeholder engagement:** Communication and trust-building
- **Crisis management:** Plan for AI-related incidents

WHY OTHER OPTIONS ARE WRONG:

- A: This IS a mitigation strategy
- C: This IS a mitigation strategy
- D: This IS a mitigation strategy

EXAM TRAP: Ignoring incidents is NOT a valid mitigation strategy!

QUESTION: What is superintelligence?

OPTIONS:

1. An AI system that is faster than humans
2. An AI system that operates beyond human-level intelligence
3. An AI system that is less intelligent than humans
4. An AI system that only performs one task

ANSWER: B

DETAILED EXPLANATION:

- AI beyond human-level intelligence
- Could act in ways threatening to humanity
- Concern: well-intentioned AI with misaligned values
- Central to existential risk discussions

WHY OTHER OPTIONS ARE WRONG:

- A: Speed is different from intelligence level
- C: Superintelligence is ABOVE human level
- D: Superintelligence would be general, not narrow

EXAM TRAP: Superintelligence = beyond human-level intelligence!

Q82: AI Alignment - Tutorial

QUESTION: What is AI alignment?

OPTIONS:

1. Aligning AI systems with human values
2. Aligning AI systems with maximum efficiency
3. Aligning AI systems with profit maximization
4. Aligning AI systems with speed

ANSWER: A

DETAILED EXPLANATION:

- Ensuring AI systems' means and goals align with human values
- Example: Paperclip maximizer scenario
- Misaligned AI could cause catastrophic harm
- Central concern for existential risk

WHY OTHER OPTIONS ARE WRONG:

- B: Efficiency is not alignment
- C: Profit is not alignment
- D: Speed is not alignment

EXAM TRAP: AI alignment = matching AI to human values!

Q83: AI Arms Race - Tutorial

QUESTION: What is the AI arms race concern?

OPTIONS:

1. Competing nations developing AI for peaceful purposes
2. Risks from different stakeholders creating powerful AI without adequate safety measures
3. A race to create the fastest AI
4. A race to create the most profitable AI

ANSWER: B

DETAILED EXPLANATION:

- Multiple stakeholders creating powerful AI
- Without adequate safety measures
- Could lead to catastrophic conflicts
- Part of existential risk discussions

WHY OTHER OPTIONS ARE WRONG:

- A: The concern is about UNSAFE development
- C: Speed is not the primary concern
- D: Profit is not the primary concern

EXAM TRAP: AI arms race = unsafe AI development competition!

Q84: Economic Risks from AI - Tutorial

QUESTION: What is a major economic risk from AI?

OPTIONS:

1. AI always creates more jobs
2. AI never affects employment
3. Displacement of jobs by AI and widening inequality
4. AI has no economic impact

ANSWER: C

DETAILED EXPLANATION:

- Jobs could be displaced by automation
- High-skill jobs also at risk (accounting, legal)
- Widening gap between AI-skilled and unskilled
- Need for workforce re-skilling
- Predictions vary (Goldman Sachs: 1/4 jobs could be automated)

WHY OTHER OPTIONS ARE WRONG:

- A: AI may DESTROY some jobs
- B: AI DOES affect employment
- D: AI HAS significant economic impact

EXAM TRAP: Economic risks = job displacement and inequality!

Q85: Global Inequality from AI - Tutorial

QUESTION: How could AI exacerbate global inequality?

OPTIONS:

1. AI is evenly distributed globally
2. Nations at the center of AI development leap ahead, leaving others behind
3. AI has no effect on global inequality
4. AI reduces inequality automatically

ANSWER: B

DETAILED EXPLANATION:

- Most large AI firms based in the US
- Technological adoption varies widely
- AI advancement could widen global inequality
- Nations in AI development leap ahead
- Others left behind in economic growth and influence

WHY OTHER OPTIONS ARE WRONG:

- A: AI is NOT evenly distributed
- C: AI CAN exacerbate inequality
- D: AI does NOT automatically reduce inequality

EXAM TRAP: AI could widen global inequality!

QUESTION: According to McKinsey (2025), what percentage of organizations use GenAI?

OPTIONS:

1. 33%
2. 71%
3. 50%
4. 95%

ANSWER: B

DETAILED EXPLANATION:

- 71% of organizations regularly use GenAI
- Up from 33% in 2023
- Particularly high in digitized industries
- Banking and financial services: 58% embrace GenAI

WHY OTHER OPTIONS ARE WRONG:

- A: This was the 2023 figure
- C: The figure is higher
- D: This is the BCG figure for professional services

EXAM TRAP: GenAI adoption = 71% of organizations (2025 McKinsey)!

Q87: Hallucination Definition - Tutorial

QUESTION: What is hallucination in GenAI?

OPTIONS:

1. Always correct outputs
2. Factually incorrect, misleading, or fabricated outputs that appear credible
3. Outputs that are never credible
4. Outputs that are always correct

ANSWER: B

DETAILED EXPLANATION:

- Generate factually incorrect output
- Appear credible to non-experts
- Problematic in legal, financial, healthcare contexts
- Can lead to reputational damage and legal liability

WHY OTHER OPTIONS ARE WRONG:

- A: Hallucinations are INCORRECT
- C: Hallucinations APPEAR credible
- D: Hallucinations are NOT correct

EXAM TRAP: Hallucination = incorrect but credible outputs!

Q88: Air Canada Chatbot Case - Tutorial

QUESTION: What happened in the Air Canada chatbot case?

OPTIONS:

1. The chatbot worked perfectly
2. The chatbot offered a discount that shouldn't have been available, and the company was held responsible
3. The chatbot was a separate legal entity
4. The company won the case

ANSWER: B

DETAILED EXPLANATION:

- Chatbot offered a passenger a discount
- Discount shouldn't have been available
- Air Canada argued chatbot was separate legal entity
- Tribunal rejected argument
- Company responsible for chatbot outputs
- Awarded damages and tribunal fees

WHY OTHER OPTIONS ARE WRONG:

- A: The chatbot made an error
- C: The argument was REJECTED
- D: Air Canada LOST the case

EXAM TRAP: Companies are liable for chatbot outputs!

Q89: Unpredictability in GenAI - Tutorial

QUESTION: Why is unpredictability a risk in GenAI?

OPTIONS:

1. GenAI outputs are always predictable
2. Unpredictability undermines standardization across users and time
3. Unpredictability is always beneficial
4. Unpredictability never affects compliance

ANSWER: B

DETAILED EXPLANATION:

- GenAI systems are unpredictable in output
- Challenges standardization across users and time
- Problems for auditability
- Affects financial advice, claims handling, legal contexts
- Poses compliance and reputational risks

WHY OTHER OPTIONS ARE WRONG:

- A: GenAI outputs are NOT always predictable
- C: Unpredictability is generally problematic
- D: Unpredictability CAN affect compliance

EXAM TRAP: Unpredictability undermines standardization and auditability!

Q90: Prompt Injection - Tutorial

QUESTION: What is prompt injection?

OPTIONS:

1. A technique to improve outputs
2. Hidden instructions within user inputs to cause unintended behavior
3. A method to increase accuracy
4. A technique to remove bias

ANSWER: B

DETAILED EXPLANATION:

- Hidden instructions in user inputs
- Cause unintended behavior
- One of several adversarial input techniques
- Can cause GenAI systems to behave unsafely

WHY OTHER OPTIONS ARE WRONG:

- A: Prompt injection causes PROBLEMS
- C: Prompt injection is about manipulation
- D: Prompt injection doesn't remove bias

EXAM TRAP: Prompt injection = hidden instructions to cause unintended behavior!

Q91: Jailbreaking Definition - Tutorial

QUESTION: What is jailbreaking in the context of GenAI?

OPTIONS:

1. Making the model faster
2. Bypassing a model's safety mechanisms through crafted prompts
3. A method to improve outputs
4. A technique to increase accuracy

ANSWER: B

DETAILED EXPLANATION:

- Crafted prompts to bypass safety mechanisms
- Override safety or policy constraints
- Can cause unintended or unsafe behavior
- Type of adversarial input

WHY OTHER OPTIONS ARE WRONG:

- A: Speed is not the purpose
- C: Jailbreaking is about bypassing safety
- D: Jailbreaking does NOT increase accuracy

EXAM TRAP: Jailbreaking = bypassing safety mechanisms!

Q92: Data Privacy Violations in GenAI - Tutorial

QUESTION: How can GenAI cause data privacy violations?

OPTIONS:

1. GenAI never accesses personal data
2. Prompts containing PII may be inadvertently stored, shared, or used for training
3. Data privacy is not a concern
4. GenAI always protects privacy

ANSWER: B

DETAILED EXPLANATION:

- User prompts may contain personally identifiable information (PII)
- Data may be inadvertently stored or shared
- May be used for model training without user knowledge
- Example: Researchers extracted emails, phone numbers from LLM
- FTC investigation into OpenAI for privacy issues

WHY OTHER OPTIONS ARE WRONG:

- A: GenAI CAN access personal data
- C: Data privacy IS a concern
- D: GenAI does NOT always protect privacy

EXAM TRAP: GenAI prompts may inadvertently expose PII!

Q93: Copyright Issues in GenAI - Tutorial

QUESTION: What copyright concern exists with GenAI?

OPTIONS:

1. GenAI never uses copyrighted material
2. GenAI models are trained on web-scraped content including copyrighted material
3. Copyright laws don't apply to AI
4. All GenAI content is public domain

ANSWER: B

DETAILED EXPLANATION:

- Trained on vast web-scraped content
- Includes copyrighted books, images, articles
- Lively debate and legal proceedings
- Example: Getty Images vs. Stability AI
- Using generated content may carry copyright risks

WHY OTHER OPTIONS ARE WRONG:

- A: GenAI DOES use copyrighted material
- C: Copyright laws DO apply
- D: GenAI content is NOT public domain

EXAM TRAP: GenAI training may infringe copyright!

Q94: Responsibility Gaps - Tutorial

QUESTION: What are responsibility gaps in GenAI adoption?

OPTIONS:

1. Clear lines of accountability always exist
2. GenAI tools may be adopted across teams without clear responsibility for oversight
3. Responsibility is never an issue
4. All employees are always responsible

ANSWER: B

DETAILED EXPLANATION:

- GenAI tools adopted by everyone
- Non-technical teams may use LLMs
- Example: HR team screening complaints
- Biased or unfair outputs possible
- Need clear lines of responsibility

WHY OTHER OPTIONS ARE WRONG:

- A: Responsibility gaps ARE a problem
- C: Responsibility IS an issue
- D: Not all employees have clear responsibility

EXAM TRAP: GenAI adoption creates responsibility gaps across teams!

Q95: Misaligned Incentives - Tutorial

QUESTION: What misalignment exists in GenAI adoption?

OPTIONS:

1. Companies always prioritize safety over speed
2. Pressure to innovate can outweigh cautionary approaches
3. Regulation always prevents problems
4. Incentives are always aligned with safety

ANSWER: B

DETAILED EXPLANATION:

- Pressure to cut costs and improve productivity
- Can outweigh safety considerations
- Example: Anthropic lobbied to change safety bill
- Teams rewarded for cost savings, not risk flagging
- Need review structures tied to responsible AI

WHY OTHER OPTIONS ARE WRONG:

- A: Safety is NOT always prioritized
- C: Regulation is not always effective
- D: Incentives are NOT always aligned

EXAM TRAP: Innovation pressure can outweigh safety considerations!

Q96: Lack of Training Risk - Tutorial

QUESTION: What risk arises from lack of GenAI training?

OPTIONS:

1. Employees always use GenAI correctly
2. Employees may use GenAI without understanding acceptable use or data protection obligations
3. Training is never needed
4. GenAI always protects data

ANSWER: B

DETAILED EXPLANATION:

- LLM chatbots are intuitive
- Employees integrate into workflows without guidance
- May paste sensitive data into ChatGPT (4.2
- Risk to data protection, reputational risk
- Companies like JPMorgan banned ChatGPT

WHY OTHER OPTIONS ARE WRONG:

- A: Employees may NOT use GenAI correctly
- C: Training IS needed
- D: GenAI does NOT always protect data

EXAM TRAP: Lack of training leads to unintentional data exposure!

Q97: Vendor Lock-In - Tutorial

QUESTION: What is vendor lock-in in GenAI?

OPTIONS:

1. Complete freedom to switch vendors
2. Dependence on external providers, limiting auditability and leverage
3. Vendors never change pricing
4. All providers are identical

ANSWER: B

DETAILED EXPLANATION:

- Most organizations rely on external providers
- Few dominant suppliers
- Dependencies create power asymmetry
- Providers may change pricing/usage without warning
- Short-term adoption $\rightarrow$ long-term structural lock-in

WHY OTHER OPTIONS ARE WRONG:

- A: Switching is often difficult
- C: Vendors CAN change terms
- D: Providers differ significantly

EXAM TRAP: Vendor lock-in = dependence limiting control!

QUESTION: What risk does synthetic content pose?

OPTIONS:

1. Synthetic content is always beneficial
2. High-quality misleading content can be created at scale, damaging trust and institutions
3. Synthetic content is easy to identify
4. Synthetic content never affects elections

ANSWER: B

DETAILED EXPLANATION:

- GenAI lowers cost of creating misleading content
- Fake news, health misinformation, political content
- Can damage democratic institutions
- Can lead to distrust of legitimate sources
- EU Code of Practice on Disinformation addresses risks

WHY OTHER OPTIONS ARE WRONG:

- A: Synthetic content CAN be harmful
- C: Synthetic content is often difficult to identify
- D: Synthetic content CAN affect elections

EXAM TRAP: Synthetic content can damage trust and institutions!

Q99: Deepfakes Risks - Tutorial

QUESTION: What risk do deepfakes pose?

OPTIONS:

1. Deepfakes are always harmless
2. Audio or video deepfakes can convincingly mimic individuals, causing fraud, defamation, or political manipulation
3. Deepfakes are easy to detect
4. Deepfakes never cause harm

ANSWER: B

DETAILED EXPLANATION:

- Convincingly mimic public figures or private individuals
- Individual and societal harm
- Fraud: CEO persuaded to transfer €220,000
- Defamation, political manipulation
- Even low-effort impersonations can go viral

WHY OTHER OPTIONS ARE WRONG:

- A: Deepfakes CAN cause harm
- C: Deepfakes are often difficult to detect
- D: Deepfakes CAN cause harm

EXAM TRAP: Deepfakes can cause fraud, defamation, and manipulation!

Q100: GenAI Risk Management Summary - Tutorial

QUESTION: What is the most comprehensive summary of GenAI risk management?

OPTIONS:

1. GenAI risks are purely technical
2. GenAI risks include functional, data-related, organizational, strategic, and societal risks, requiring integrated oversight and governance
3. GenAI risks are not significant
4. Only hallucinations matter

ANSWER: B

DETAILED EXPLANATION:

- **Functional:** Hallucinations, unpredictability, adversarial inputs
- **Data-related:** Privacy violations, copyright issues
- **Organizational:** Responsibility gaps, misaligned incentives, lack of training
- **Strategic:** Vendor lock-in, regulatory unpredictability
- **Societal:** Disinformation, deepfakes, manipulation

WHY OTHER OPTIONS ARE WRONG:

- A: Risks extend beyond technical
- C: GenAI risks ARE significant
- D: Multiple risk categories exist

EXAM SUCCESS: Understand all risk categories and their interconnections!